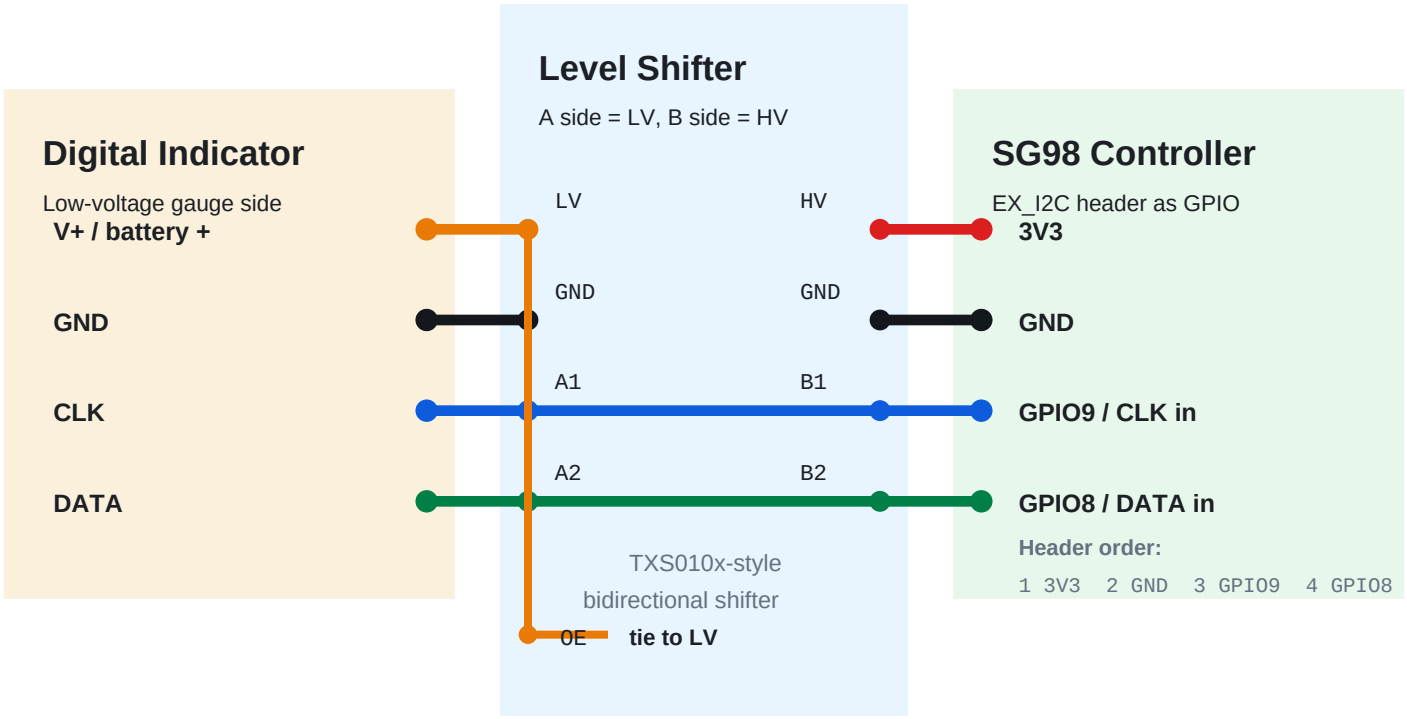
SG98 Burr-Gap Digital Indicator Wiring

GUIRUIMOON FS PFC 12mm indicator -> 4-channel level shifter -> SG98 controller EX_I2C header

Assumption: the indicator exposes the common 4-wire clock/data port: V+, GND, CLK, DATA.
Keep the indicator battery installed unless you confirm the gauge accepts external 3.3V power.



Connection Table

Indicator	Level shifter	Controller
V+ / battery +	LV and OE	no direct controller pin
GND	GND	EX_I2C pin 2 / GND
CLK	A1 -> B1	EX_I2C pin 3 / GPIO9
DATA	A2 -> B2	EX_I2C pin 4 / GPIO8
none	HV	EX_I2C pin 1 / 3V3

Bring-up notes

1. Confirm GND and V+ at the gauge battery/data connector before wiring.
2. Probe the remaining two pins while moving the plunger; assign CLK and DATA by waveform.
3. Do not connect gauge CLK/DATA directly to ESP32 GPIO. Use the level shifter.
4. **EX_I2C is only used as free GPIO here; this is not an I2C device.**